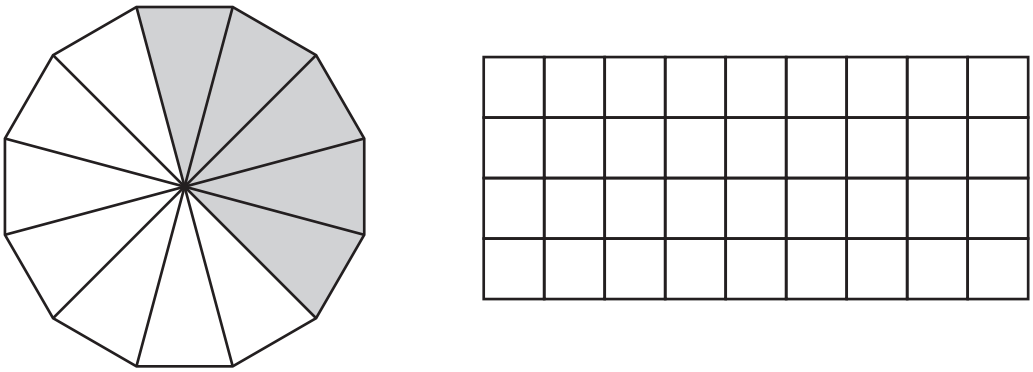
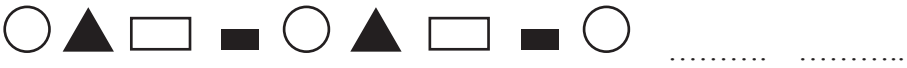
0580_s22_qp_32_Q7

7 **(a)**



Shade some squares so that both shapes have the same fraction shaded. [2]

(b) Here is a pattern.



Position number 1 is a ○.

Position number 2 is a ▲.

(i) Draw the next two shapes in this pattern. [1]

(ii) What do the position numbers of the shape ▬ have in common?

..... [1]

(iii) Pierre says that the shape in position number 99 is a ▭.

Explain why he is correct.

.....

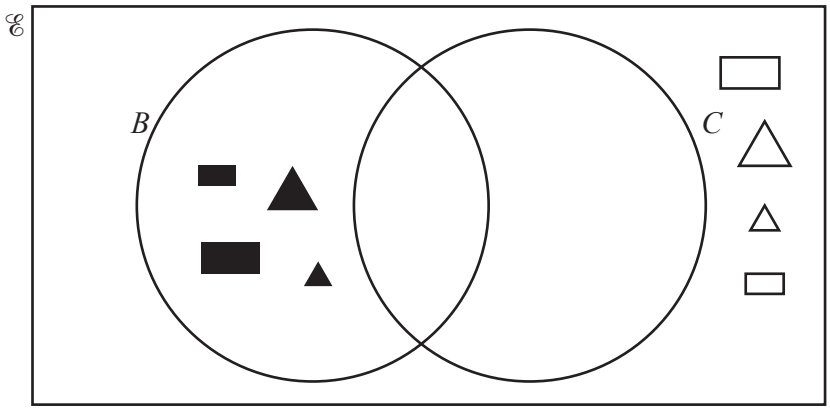
..... [2]

(c) $\mathscr{E}$ = { ●, ○, •, ∘, ▲, △, ▲, △, ▬, ▭, ▬, ▭ }

This universal set has twelve elements.
Each shape is:

- a circle, C, or a triangle, T, or a rectangle, R
- large, L, or small, S
- black, B, or white, W.

(i)



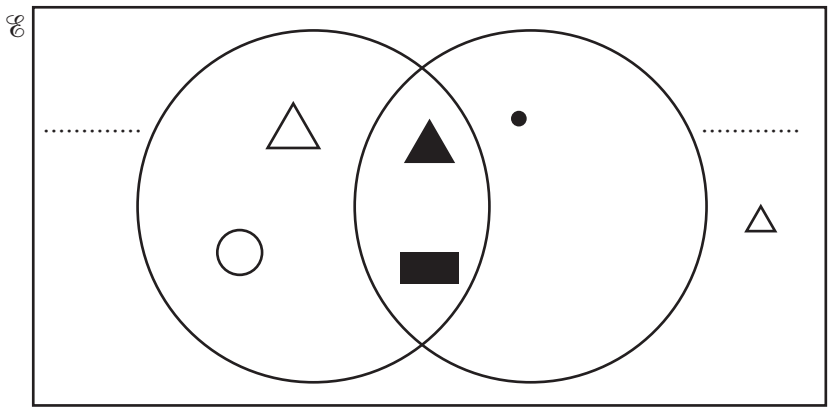
The triangles and rectangles are drawn in the Venn diagram.

(a) Draw the four circles to complete the Venn diagram. [1]

(b) Find $n(B \cup C)$.

..... [1]

(ii) Six of the twelve shapes are drawn in another Venn diagram.



Complete the Venn diagram by:

- labelling the sets

and

- drawing the shapes ●, ○, ▲, ▭, ▬ and ▭ . [3]